

**COURSE TITLE : BASICS OF COMPUTER HARDWARE AND NETWORKING**  
**COURSE CODE : 3259**  
**SEMESTER : 3**  
**COURSE CATEGORY : B**  
**PERIODS/WEEK : 4**  
**PERIODS/SEMESTER : 60**  
**CREDITS : 2**

#### **TIME SCHEDULE**

| <b>TOPICS</b>                        | <b>PERIODS</b> |
|--------------------------------------|----------------|
| VARIOUS EXERCISES (Exercise 1 to 8)  | <b>20</b>      |
| VARIOUS EXERCISES (Exercise 9 to 15) | <b>20</b>      |
| Trouble shooting                     | <b>10</b>      |
| Networking                           | <b>10</b>      |

#### **Course General Outcomes:**

| <b>Sl.</b> | <b>G.O</b> | <b>On completion of this course the student will be able :</b> |
|------------|------------|----------------------------------------------------------------|
| <b>1</b>   | <b>1</b>   | To understand various hardware components                      |
| <b>2</b>   | <b>1</b>   | To know about motherboard, monitor, keyboard etc.              |
| <b>3</b>   | <b>1</b>   | To know about installation of Operating System and softwares   |
| <b>4</b>   | <b>1</b>   | To understand trouble shooting & networking                    |

#### **SPECIFIC OUTCOMES**

##### **LIST OF EXERCISES**

1. Introduction to Fundamentals of Computers -Types of Computers-Machine-assembly and High level Languages- - Data Organization-Types of memory-Secondary storage Devices-I/O devices.
2. Demonstration of mother board – identification of processor sockets and CPU support, memory module support, power connector, keyboard connector, BIOS, bus slots, chipset, cache memory, co-processors, CMOS battery, CMOS chipset etc
3. Demonstration of SMPS – connector identification - plugging the connector in motherboards and devices like HDDs, CD-ROMs, etc – various power supply problems.
4. Demonstration of various monitors – CRT, TFT, LCD
5. Demonstration of keyboards – identifying various types of keyboards
6. Demonstration of HDDs – identifying the HDDs with different interfacing – connecting details – master slave connection
7. Demonstrating the system assembly – CPU – cooler fan, SMPs – memory monitor – keyboard – HDD
8. Demonstrating the errors during the POST – configuring the peripherals in CMOS RAM
9. Demonstration of fixing the parts in the cabinet - connecting front panel connections

10. Demonstration of various with serial – parallel – PS-2 interfaces – USB ports
11. Demonstration of CDs – Drives – Drive installation procedure – CD writer installation and writing practice
12. Installation windows and Linux operating system from CDs and HDDs – installation of drivers of display
13. Demonstration of Laser Printers – Dot Matrix Printers – Inkjet Printers – Installation – various troubles with printers
14. Demonstration of scanner – installation and connection procedures
15. Installation of Various Application Software

### **Trouble Shooting**

1. Various troubles and rectification procedures of motherboard – CPU – memory modules – monitors – SMPS – keyboards, HDD – CMOS setup – CD ROM driver – Printers– scanners – modern - mouse
2. OS related problems (WINDOWS &, LINUX)
3. Virus problem and solution – Antivirus software

### **Networking**

1. Demonstration of networking – Demonstration of Network Components – Switches – Routers Internet Modems – UTP Cable Crimping - network administration – IP Addressing - user management – printer connection – basic troubleshooting on networks
2. Internet – Intranet - E-mail - Internet Sharing – Network Printers – Trouble Shooting

### **BOOKS RECOMMENDED**

1. “Comdex Hardware & Networking Course Kit” by Gupta
2. “Trouble shooting maintaining & Repairing of PCs” by Bigelow by Tata
3. “Real World Networking Trouble Shooting Manual” by Alen Sugan